

8DM60: Multimodal AI in Biomedicine

Lecture 6: Guided self-study (BZ)

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Problem 1

A developer adds the name, description, and input format of a nucleus detector to an LLM's prompt. The model now produces correctly formatted requests for the detector. The developer claims that the integration is complete.

Is this sufficient to establish that the integration works? Justify your assessment.

Problem 2

A laboratory has a validated analysis procedure: check image quality, obtain specified measurements, and prepare a standard report. Poor-quality images require replacement; incomplete measurements require human review. These conditions and their responses are known in advance.

A developer proposes an agent that chooses each next step, arguing that the exceptions make a fixed workflow unsuitable.

Recommend either a predefined workflow or an agent for this application. Base your justification on what determines the possible paths, rather than whether the system contains branches or uses an LLM.

Problem 3

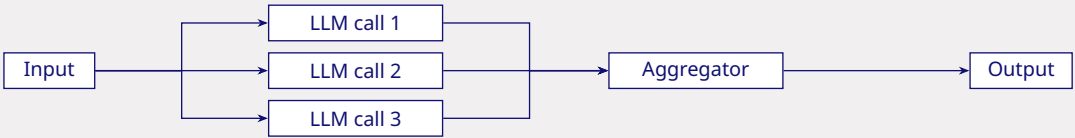
Two AI assistants for clinical use are being designed:

Assistant	Requirement
A	Prepare a consultation by separately summarising the medication list, recent blood tests, and imaging reports. These three summaries can be produced independently before being combined.
B	Support a clinician reviewing a complex case. The presenting problem and initial findings determine which additional record reviews and specialist analyses are relevant.

Choose between predefined parallelization and orchestrator–workers for each assistant. Explain the architectural difference that matters here and the extra coordination cost that must be justified when subtasks are assigned dynamically.

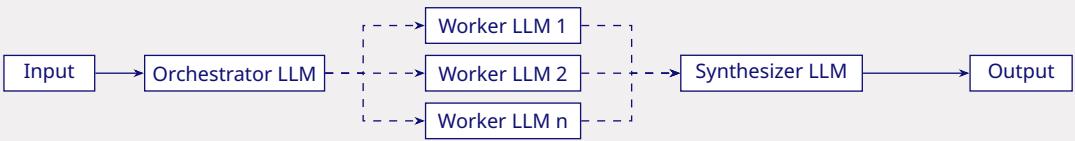
Problem 3: architectures

Predefined parallelization



Subtasks are defined in advance and run independently.

Orchestrator–workers



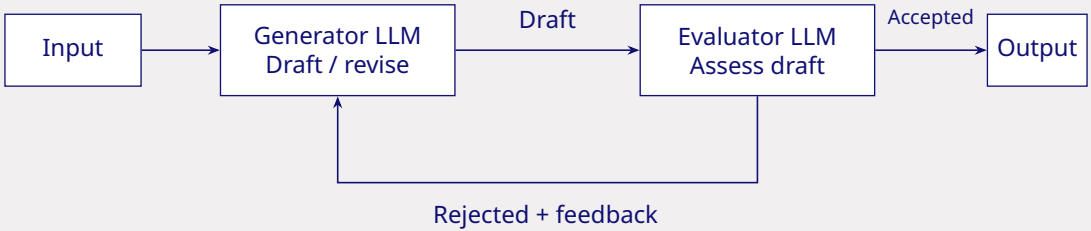
The orchestrator defines and assigns subtasks for the case.

Problem 4

A system drafts patient summaries from clinical notes, laboratory results, and vital-sign measurements. Each summary must include the requested findings, preserve values, units and chronology, and be clear to a clinician. The team is considering an evaluator–optimizer workflow to improve drafts before release.

Propose the evaluation and stopping criteria for this workflow. Identify which criteria can be checked directly against the source records and which require a judgement of the text. Your design should distinguish a draft that passes the checks from a run that merely reaches its revision limit.

Problem 4: evaluator–optimizer



The evaluator uses the source records and evaluation criteria.

Problem 5

A hospital is considering replacing a predefined patient-record review workflow with an agent. Both use the same LLM and the same tools for retrieving clinical notes, laboratory results, and vital-sign histories. When all relevant information is supplied, their case summaries are similarly accurate. The proposed benefit of the agent is its ability to identify missing information and resolve inconsistencies between records.

Design an AI-SCE-style simulated evaluation that could justify either keeping the workflow or adopting the agent. Specify the situations to include, the interactions to record, and how to compare outcomes, process quality, and resource use. The evaluation must be able to distinguish a well-supported case summary from an accidentally correct one.